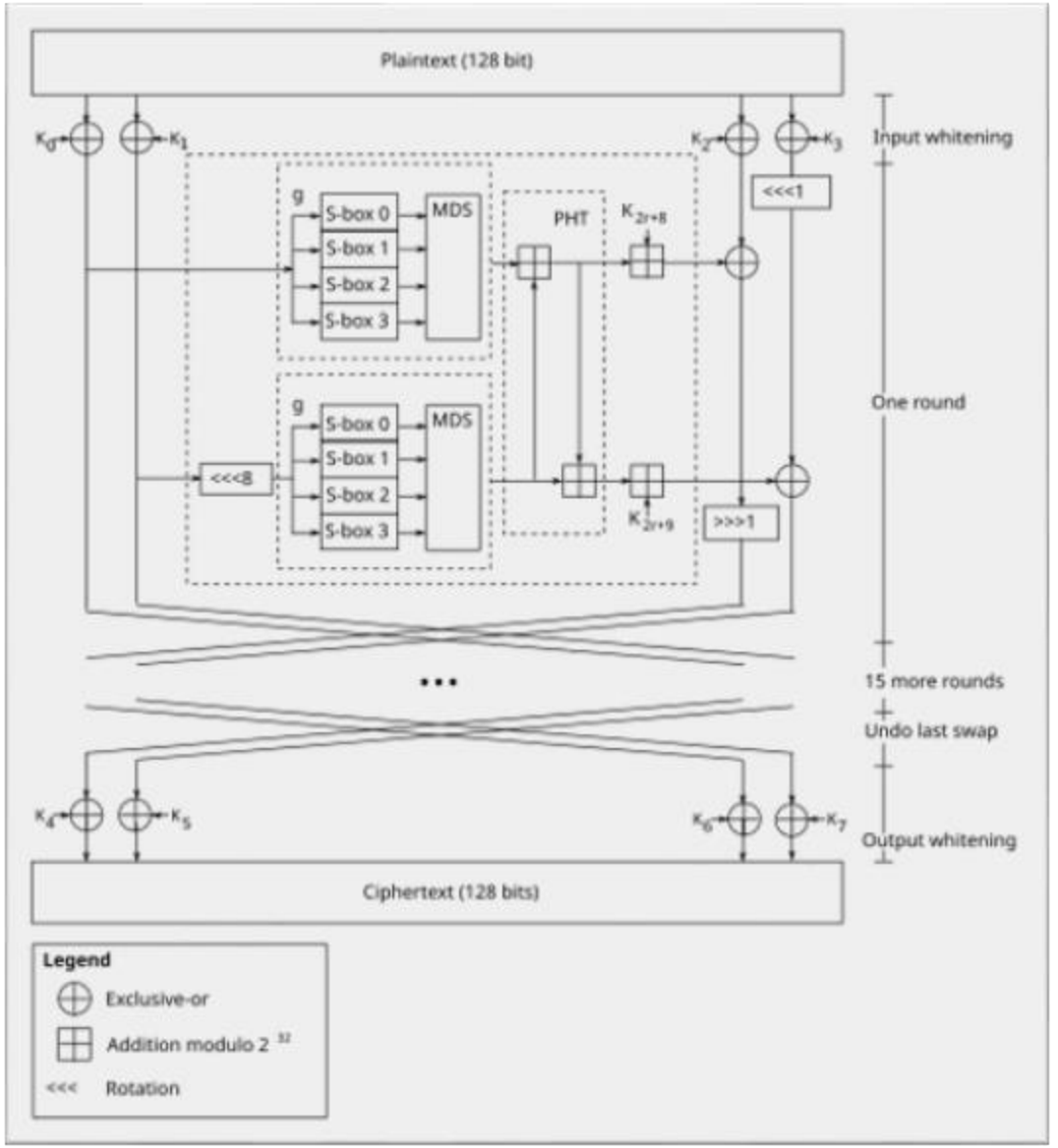
TwoFish 128-Bit Encryption Implementation



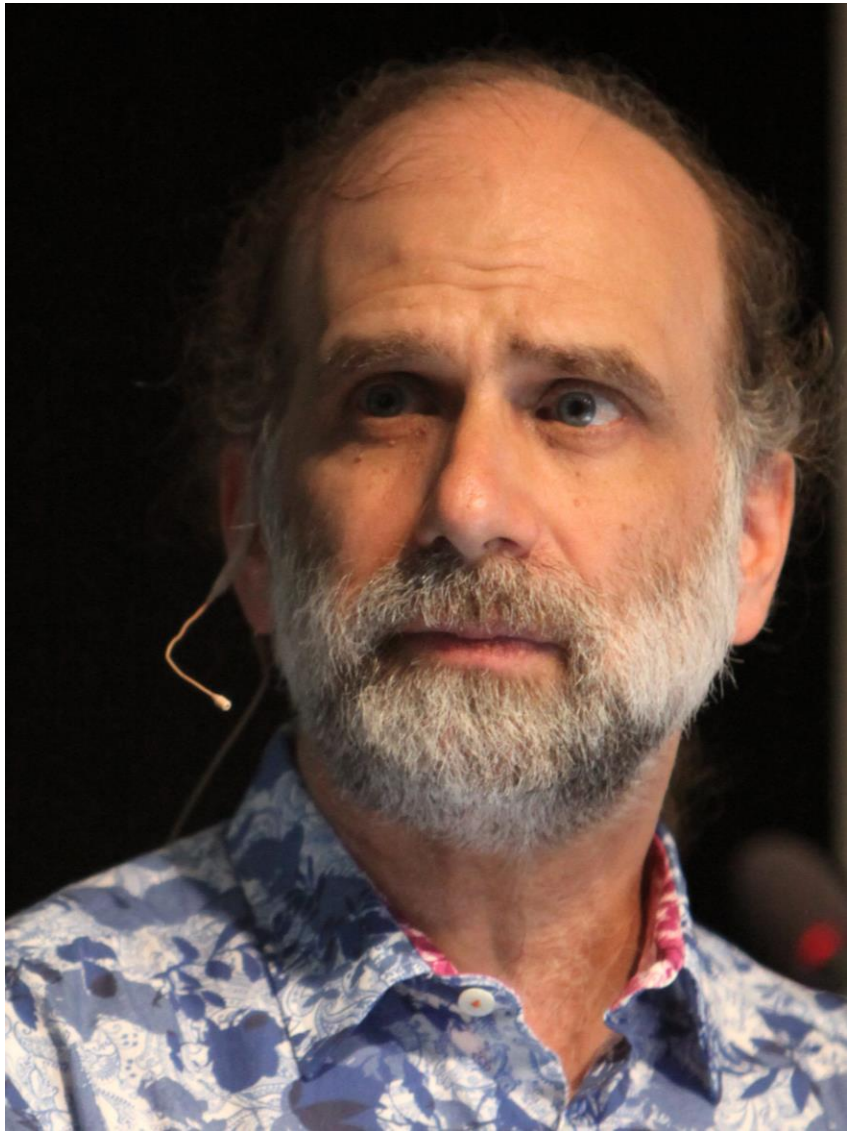
Tallen Lawrence

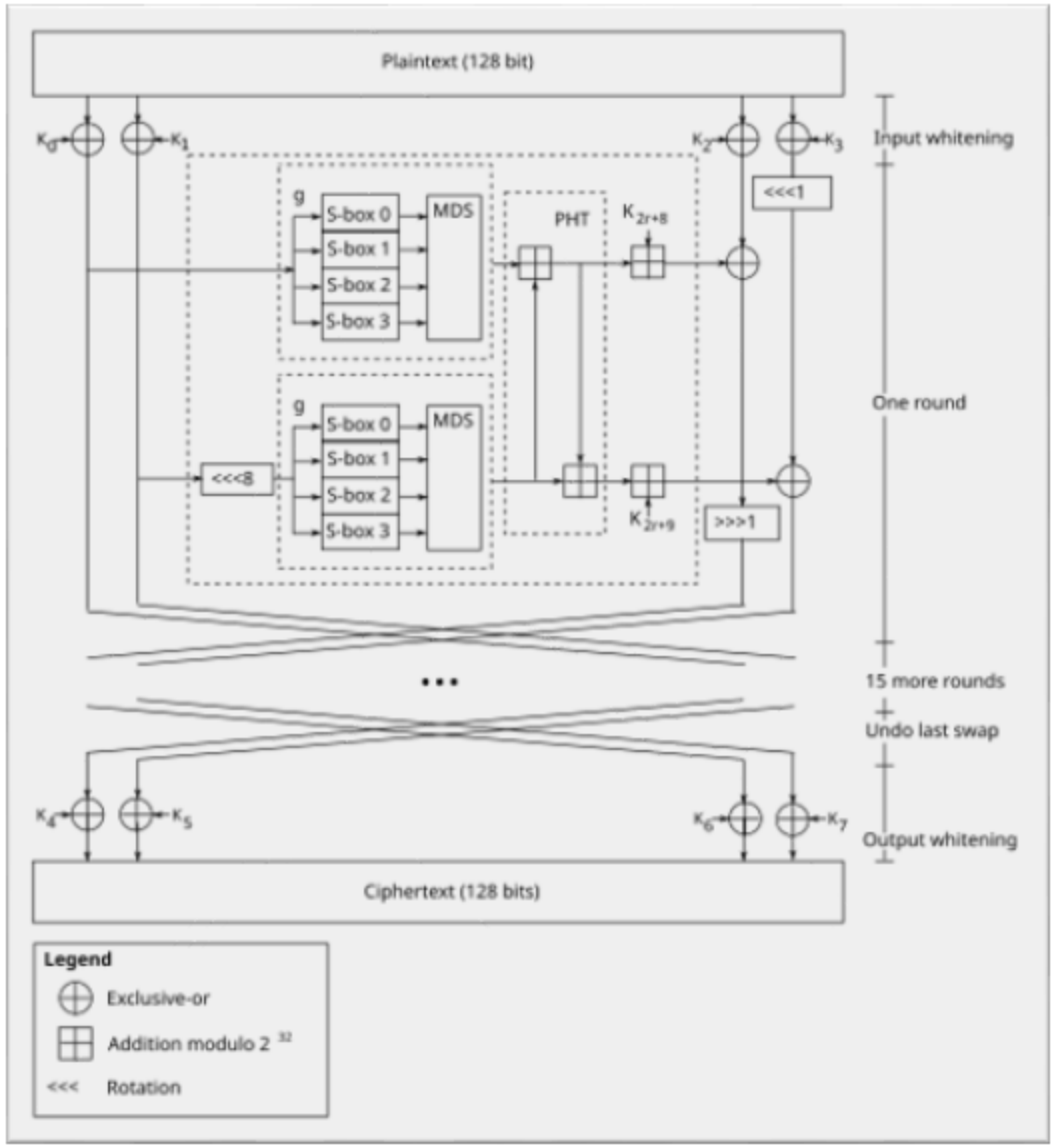
CS:4660 Cryptography-Spring 2026



Brief Background

- Developed by Bruce Schneier and Counterpane Labs, published in 1998
- Was one of five Advanced Encryption Standard (AES) finalists, but ultimately lost
- Derived from Blowfish-64 cipher
- Highly secure, but considered slower than Rijndael





Overview

- TwoFish is symmetric key block cipher
- 128-bit block size, variable key size of up to 256-bits
- Follows a 16-round Feistel network structure with key-dependent S-boxes
- Complex Key Schedule

Key Schedule

- Generates subkeys S0 and S1 (used for function g)
- Function h (on right) generates 40 subkeys k
  - K0-k8 used for input whitening
  - K9-k30 used for each round of encryption
  - K31-39 used for output whitening

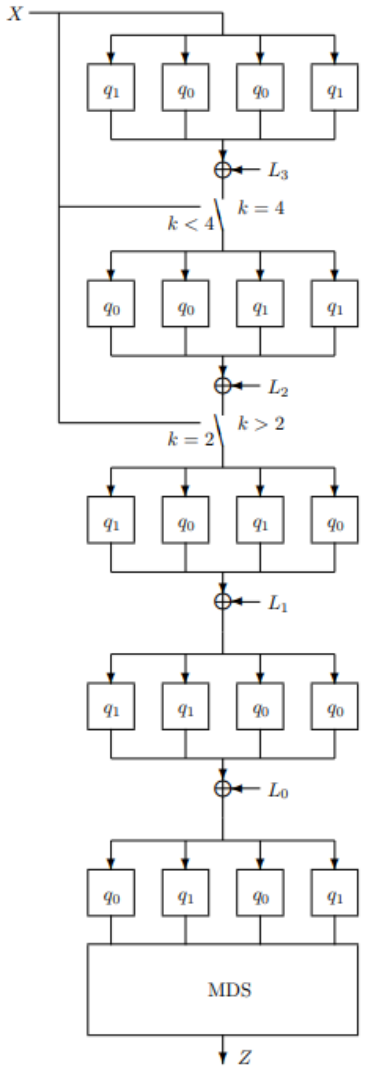


Figure 2: The function h

Function F

- Two 32-bit blocks (R0 and R1) passed through function g
- The outputs then combined in Pseudo Hadamard Transform (PHT)
- Outputs then combined with 32-bit round keys generated by key schedule to form 128-bit word.

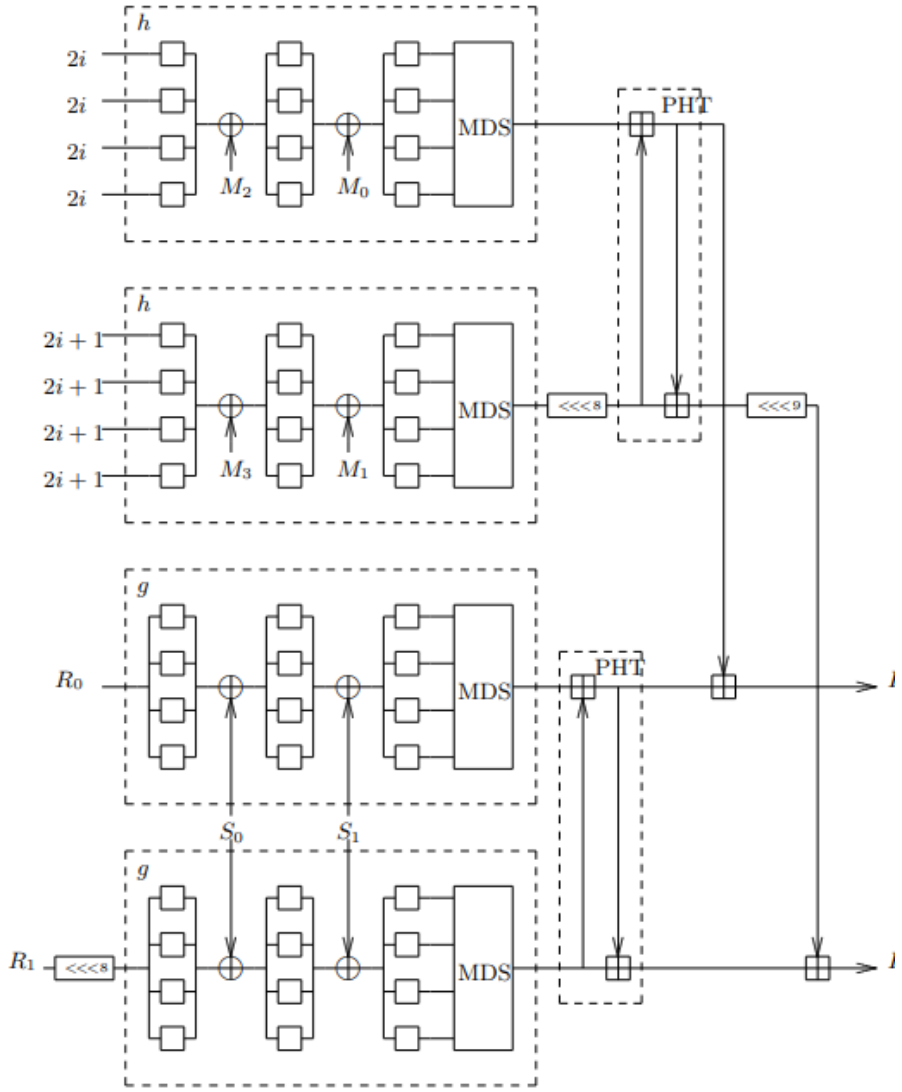
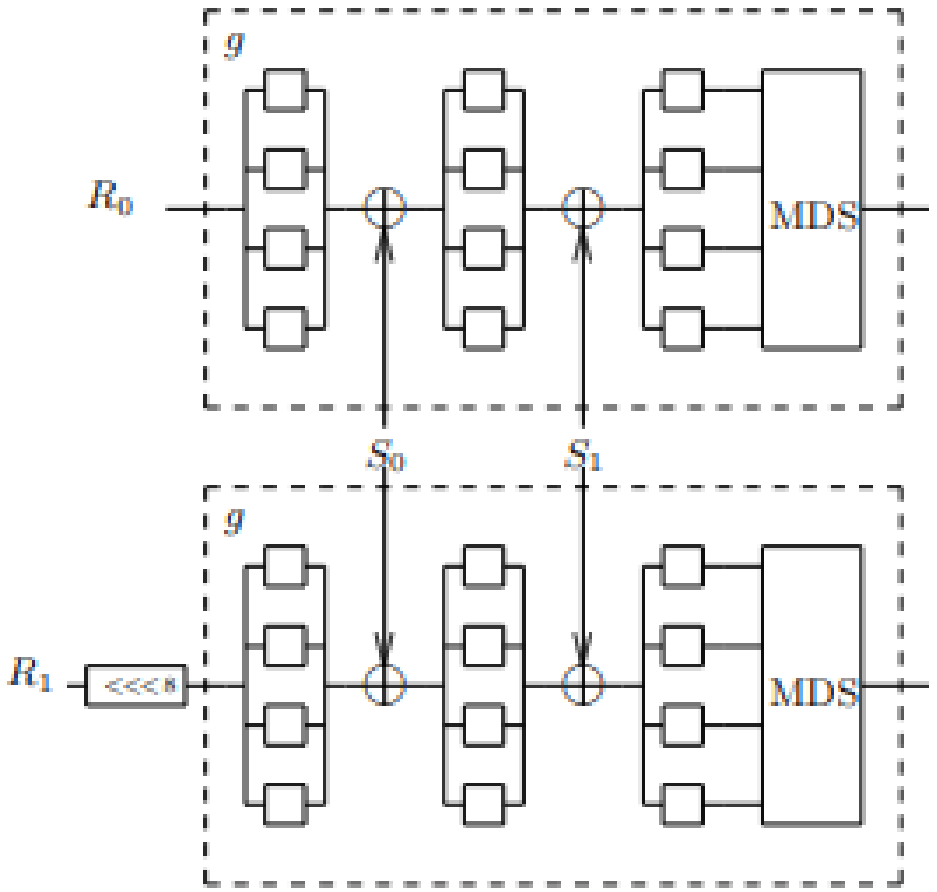


Figure 3: A view of a single round F function (128-bit key)



Function g (S box)

- 32-bit input split into four 8-bit nibbles
- 3 Rounds of q permutation (q0 for R0, q1 for R1) broken up by two rounds of XOR with subkeys S0 and S1
- 8-bit nibbles inputted into Maximum Distance Separate (MDS) matrix, final output 32-bit

Demo